

Erlang Age-Transport Dynamics: Algebraic Renewal Roots versus the Essential Spectral Edge

Route-A source-local certificate HCS-C272

1 September 2026

Abstract

We solve an Erlang-fertility McKendrick age-transport semigroup at the level of characteristics, renewal poles, and long-time dynamics. The Euler–Lotka denominator has exactly k explicit algebraic roots. No renewal pole is promoted to target arithmetic data.

1 Erlang age-transport owner

On $X = L^1(\mathbb{R}_+, da)$, freeze

$$\partial_t n + \partial_a n = -\mu n, \quad n(t, 0) = \int_0^\infty b_k(a) n(t, a) da, \quad (1)$$

$$b_k(a) = \frac{\beta \gamma^k a^{k-1} e^{-\gamma a}}{(k-1)!}, \quad \mu, \gamma, \beta > 0, \quad k \geq 1. \quad (2)$$

Age-structured transport descends from M’Kendrick’s mathematical population framework [1]; the Erlang specialization and spectral audit are local.

Characteristics transport initial mass to the right with factor $e^{-\mu t}$ and fill the newborn triangle from the boundary. Substitution into (1) produces a scalar renewal equation. Its exponential mode has denominator

$$\Delta(\lambda) = 1 - \beta \left(\frac{\gamma}{\gamma + \lambda + \mu} \right)^k. \quad (3)$$

Therefore the algebraic roots are

$$\lambda_j = \gamma \beta^{1/k} e^{2\pi i j/k} - \gamma - \mu, \quad 0 \leq j < k. \quad (4)$$

References

- [1] A. G. M’Kendrick, *Applications of Mathematics to Medical Problems*, Proceedings of the Edinburgh Mathematical Society 44 (1925), 98–130, [doi:10.1017/S0013091500034428](https://doi.org/10.1017/S0013091500034428).